Code No: 132AC **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech I Year II Semester Examinations, March/April - 2023
MATHEMATICS - III
(Common to CE, EEE, ME, ECE, CSE, EIE, IT, MCT, MMT, AE, MIE, PTM)
Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a) Define Probability Mass function and Probability Density function. [2]
b) If X is a continuous random variable with probability function $f(X=x) = ke^{-2x}$, $x>0$, then find k. [3]
c) Define Point Estimation and Interval Estimation. [2]
d) Write the assumptions of t- test? [3]
e) Define Null Hypothesis and Alternative Hypothesis. [2]
f) Define type I and type II error. [3]
g) Explain the iteration method to solve an equation. [2]
h) Write a normal equations to fit a second degree curve of the form $y = a + bx + cx^2$ for the given data. [3]
i) Using Euler's method to find y(0.2), given that $\frac{dy}{dx} = x^2 + y^2, y(0) = 1.$ [2]
j) Derive a formula to find the cube root of N using Newton-Raphson method. [3]

PART - B

(50 Marks)

2.a) The probability distribution function of X is

X	0	1	2	3	4	5	6
P(X)	K	3k	5k	K	K	11k	13k

(i) Find k (ii) Find $P(X<4), P(X\geq 5), P(3<X\leq 6).$
b) Derive moment generating function of Binomial distributions hence find mean and variance. [5+5]

OR

3.a) X is a normal variate with mean 30 and S.D 5, find the probabilities for
(i) $26\leq X\leq 40$ (ii) $X\geq 45$ (iii) $|X-30|>5.$
b) A manufacturer of cotter pins knows that 5% of his product is defective. Pins are sold in boxes of 100. He guarantees that not more than 10 pins will be defective. Determine the probability that a box will fail to meet the guarantee. [5+5]

4.a) i) Find the value of $P(-t_{0.025} < T < t_{0.05})$.

ii) Find 'k' such that $P(k < T < -1.761) = 0.045$.

b) Explain the test procedure for equality of two means in small sample tests. [5+5]

OR

5.a) Two treatments have applied on 500 agricultural plots and the data given below. Test whether treatments are independent?

Treatment - I	Treatment - II	
	208	92
	32	168

b) A random sample of size 81 was taken whose variance is 20.25 and means is 32, construct 98% confidence interval. [5+5]

6.a) To test the significance of variation in the retail prices of a commodity in three principal cities, Mumbai, Kolkata and Delhi four shops were chosen at random in each city and theprices who lack confidence in their mathematical ability observed in rupees were as follows:

Mumbai	16	8	12	14
Kolkata	14	10	10	6
Delhi	4	10	8	8

Do the data indicate that the price in the three cities are significantly different?

b) In a sample of 1000 people in Maharashtra 540 are rice eaters and the rest are wheat Eaters. Can we assume that both rice and wheat are equally popular in Maharashtra at 1% level of Significance? [5+5]

OR

7.a) Write a test procedure to test the equality of two proportions?

b) Describe about ANOVA one way classification. [5+5]

8.a) By using bisection method, find an approximate root of the equation $\sin x = \frac{1}{x}$ that lies between $x = 1$ *and* $x = 1.5$.

b) Solve by Gauss-Seidel method, the equations $20x + y - 2z = 17$, $3x + 20y - z = -18$, $2x - 3y + 20z = 25$. [5+5]

OR

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9.a) Fit a second degree parabola to the following data.

X	1.0	1.5	2.0	2.5	3.0	3.5	4.0
Y	1.1	1.3	1.6	2.0	2.7	3.4	4.1

b) An experiment gave the following vales:

v (ft/mn)	350	400	500	600
t(mn)	61	26	7	26

It's known that v and t are connected the relation $v = at^b$. Find the best possible vales of a and b. [5+5]

10.a) Evaluate $\int_0^1 \frac{dx}{1+x^2}$ by using:

i) Trapezoidal rule taking h = 1/4.
ii) Simpson's 1/3 rule taking h = 1/4.
iii) Simpson's 3/8 rule taking h = 1/6.

b) Given that $\frac{dy}{dx} = 1 + xy$ and y (0) =1, Compute y (0.1) and y (0.2) using Picard's Method.

[5+5]

OR

11. Solve the Differential equation $\frac{dy}{dx} = x + y^2$, y(0)=1 by Runge-Kutta method and compute y(0.1), y(0.2) and y(0.3). [10]

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